Signed word bridge: source-derived invariant

State: signs ±1, canonical offsets/anchors, prime 2 < r ≤ 2^63
ID initializes it; recursive find and both union orientations preserve it

Sign table closes all i8 arithmetic; existing Ring proofs close scalar operations
cu+cv ∈ {-2,0,2}; s=-cu·cv ∈ {-1,1}; no signed arithmetic overflow

Odd r = 2h+1: inverse(+2)=h+1; inverse(-2)=h
Reuse checked modular exponentiation + canonical inverse uniqueness

Invalid sign0: edge(x,0,x,1,1,0), r=101, true x=1 → stored anchor50

Fixture modulus bound is entry-point-specific; counters/index validity remain separate

Written derivation and source audit only; no new Lean check, execution or timing